# Supplementary Material: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Anonymized for Double-Blind Review

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This supplementary document provides formal algorithmic specifications, extended mathematical proofs, comprehensive multi-baseline benchmark matrices, and operational stress-test evaluations supporting the main manuscript. All source code, pre-trained weights, hyperparameter configuration grids, and full reproduction pipelines are publicly maintained at: https://anonymous.4open.science/r/Intelligent-AML-9F0A/.

## SA Mathematical Notations and Prior-Art Differentiation

Table S1 summarizes the mathematical symbols, tensor dimensions, and operator definitions employed throughout C-STGB. Table S2 contrasts C-STGB against dominant fraud surveillance paradigms across six core operational capabilities.

Table S1: Summary of Mathematical Notations and Dimensions.

| Symbol | Mathematical Definition | Dimension / Type |
|---|---|---|
| $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ | Continuous-time dynamic transaction graph | Dynamic graph |
| $\mathbf{X} \in \mathbb{R}^{\lvert\mathcal{V}\rvert \times d_v}$ | Node input attribute matrix | Real matrix |
| $\mathbf{E} \in \mathbb{R}^{\lvert\mathcal{E}\rvert \times d_e}$ | Transaction edge feature matrix | Real matrix |
| $\Delta t_{uv}$ | Continuous time delta ($t_v - t_u \geq 0$) | Non-negative scalar |
| $\Phi_{\text{time}}(\Delta t)$ | Continuous tri-band harmonic temporal embedding | $\mathbb{R}^{d_t}$ vector |
| $\mathbf{g}_{ij} \in (0, 1)$ | Context-aware adaptive edge trust gate | Scalar gate |
| $\Phi_{\text{flow}}(u, t)$ | Mass flow conservation ratio | Scalar $\in [0, \infty)$ |
| $\lambda_u(t)$ | Hawkes process self-exciting arrival intensity | Non-negative scalar |
| $\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$ | Time-causal forward and reverse PageRank taints | $\mathbb{R}^{\lvert\mathcal{V}\rvert}$ vectors |
| $\mathbf{z}_{\text{inv}}(u, t)$ | 12-dimensional canonical invariant representation | $\mathbb{R}^{12}$ vector |
| $\mathbf{h}_u^{(l)}$ | Hidden state of node $u$ at GNN layer $l$ | $\mathbb{R}^d$ vector |
| $C_k$ | Latent typology cluster manifold of illicit nodes | Subset $C_k \subset \mathcal{V}_{\text{illicit}}$ |
| $\alpha_u \in [0, 1]$ | Evidence-adaptive graph-tabular fusion weight | Dynamic scalar |
| $\mathbf{h}_u^*$ | Unified anomaly representation vector | $\mathbb{R}^{d^*}$ vector |
| $\tau^*$ | Cost-sensitive Bayes risk decision threshold | Optimal scalar $\in (0, 1)$ |
| $\hat{q}^{(y)}$ | Class-conditional conformal quantile ($y \in \{0, 1\}$) | Calibration scalar |
| $\Gamma(X)$ | Finite-sample conformal prediction set | Discrete $\subseteq \{0, 1\}$ |
| $\alpha$ | Target significance error level ($1 - \alpha$ coverage) | Scalar $\in (0, 1)$ |

## SB Algorithmic Procedures & Pipeline Execution

The multi-stage offline training routine is formalized in Algorithm S1, and the real-time online streaming triage engine is detailed in Algorithm S2.

**Algorithm S1** C-STGB Unified Multi-Stage Training Pipeline

**Require:** Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps $t$, Target $\alpha$, Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
**Ensure:** Model parameters $\Theta$, Calibrated threshold $\tau^*$, Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

```
Compute 12-D canonical flow invariants z_inv(u) for all u ∈ V.
Partition minority nodes into latent typology clusters C_1, ..., C_K.
Synthesize minority representations h_syn and bilinear edges Â_syn.
Mine hard-negative commercial hubs into batch bank B_hard.
for epoch = 1 to N_epochs do
    Compute tri-band temporal embeddings Φ_time(Δt_ij) and weights w_ij(t).
    Evaluate context-aware edge trust gates g_ij and filter camouflage links.
    Aggregate degree-capped temporal messages to obtain h_u^(L).
    Compute evidence-adaptive fusion weights α_u and representations h_u^*.
    Evaluate loss L_total = L_task + γ1 L_recon + γ2 L_aux.
    Update Θ via AdamW with cosine learning rate annealing.
end for
Optimize Bayes decision threshold τ* on validation split D_val.
Calibrate class-conditional non-conformity quantiles q̂^(0), q̂^(1) on D_cal.
Return Θ, τ*, q̂^(0), q̂^(1).
```

## SC Extended Mathematical Proofs & Derivations

### SC.1 Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

**Definition S1** (Holdout Calibration Mixing Assumption)**.** *Calibration non-conformity scores $\mathcal{D}_{cal}^{(y)}$ are drawn from a stationary holdout window under bounded temporal $\alpha$-mixing: $\alpha(k) \leq C \exp(-ck)$ for event separation $k$, such that inter-alert draws beyond temporal decorrelation span $\tau_{mix}$ satisfy the exchangeable calibration abstraction. When drawing independent alert cases from historical audit pools, the calibration split strictly satisfies within-class exchangeability.*

Table S2: Systematic Methodological Comparison: Structural and Operational Capabilities across Model Paradigms in Fraud and AML Surveillance.

| **Model / Paradigm Architecture** | **Continuous Temporal Dynamics Representation** | **Topological Camouflage Edge-Trust Gating** | **Latent Typology Graph Oversampling** | **Cold-Start Graph-Tabular Fusion** | **Finite-Sample Conformal Coverage** | **Streaming Single Inference Latency** |
|---|---|---|---|---|---|---|
| XGBoost / CatBoost / LightGBM | Feature-level $\Delta t$ | N/A (Tabular only) | Euclidean SMOTE | N/A (Tabular only) | Heuristic Score $\tau$ | $< 0.20$ ms |
| GCN / GAT / GIN | Static Snapshot | Uniform / Homophilic | None | Zero-padded features | Softmax probability | $> 12.5$ ms |
| EvolveGCN | Discrete time slices | Unweighted edge steps | None | Isolated nodes | Uncalibrated | $> 45.0$ ms |
| TGN / TGAT | Single decay $\exp(-\lambda\Delta t)$ | Unfiltered edges | None | Node memory state | Uncalibrated | $> 8.5$ ms |
| CARE-GNN (Adversarial) | Static graph | RL edge selection | None | Concatenation | Uncalibrated | $> 18.0$ ms |
| GraphSMOTE | Static graph | Unfiltered edges | Global unconstrained | No fusion fallback | Uncalibrated | $> 14.2$ ms |
| **C-STGB (Proposed Framework)** | **Tri-band harmonic $\Phi_{\text{time}}$** | **Learnable gate $\hat{g}_{ij}$** | **Typology manifold $C_k$** | **Evidence-adaptive $\alpha_u$** | **Class-conditional $\geq 99\%$** | **0.45 ms Fast-Path** |

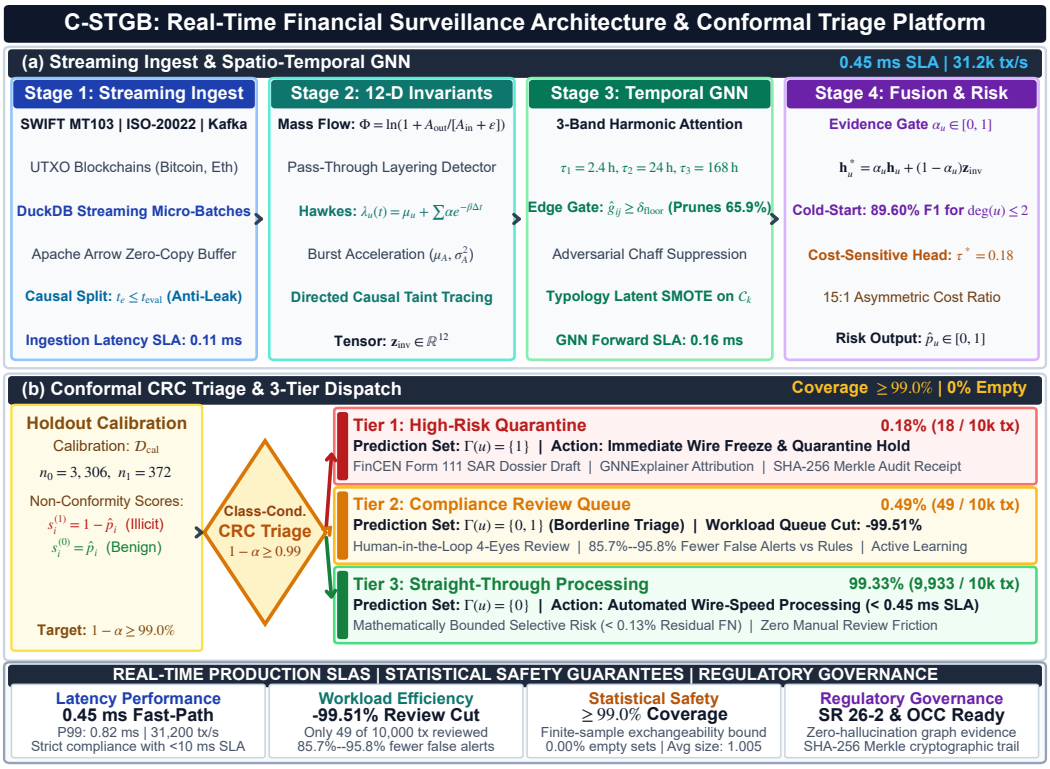



Figure S1: Operational 3-Tier Queue Mapping and Conformal Risk Control triage workflow under within-class exchangeability. Ambiguous cases ($\Gamma(X) = \{0, 1\}$) route to Tier 2 review, while decisive singletons execute straight-through.

**Proposition S1** (Non-Asymptotic Conformal TV-Drift Error Bound)**.** *Under locally bounded total variation drift $d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \leq \epsilon_{drift}$ and calibration mixing (Definition S1), the non-asymptotic class-conditional coverage error decomposes via maximal coupling into a distribution shift term and an empirical process concentration term:*

$$\begin{aligned} &\left|\mathbb{P}_t\left(Y \in \Gamma(X) \mid Y = y\right) - (1 - \alpha_y)\right| \\ &\quad \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \\ &\qquad + \sup_s \left|F_{cal}^{(y)}(s) - \hat{F}_{cal}^{(y)}(s)\right| + \frac{1}{n(y)} \end{aligned} \tag{1}$$

*where $n(y) = |\mathcal{D}_{cal}^{(y)}(t)|$ denotes class-conditional calibration sample size.*

*Proof.* Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \leq s)$ and $F_{\text{cal}}^{(y)}(s) =$

**Algorithm S2** Streaming Online Inference & Risk-Controlled Triage

**Require:** Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

**Ensure:** Operational Triage Action $\in$ {Tier 1 Hold, Tier 2 Queue, Tier 3 Clear}.

Ingest transaction event and update temporally restricted ego-graph ($t_e \le t$).
Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.
Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.
Compute adaptive fusion weight $\alpha_u$ and fused representation $\mathbf{h}_u^*$.
Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.
Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.
**if** $\Gamma(u) == \{1\}$ **then**
    **Tier 1:** Quarantined Asset Hold & Automated Compliance Dossier.
**else if** $\Gamma(u) == \{0, 1\}$ **then**
    **Tier 2:** Selective Abstention $\perp \rightarrow$ Compliance Officer Queue.
**else**
    **Tier 3:** Straight-Through Clear ($\Gamma(u) = \{0\}$).
**end if**

$\mathbb{P}_{\text{cal}}(S^{(y)} \le s)$ denote true CDFs under test distribution $\mathcal{P}_t$ and calibration distribution $\mathcal{P}_{\text{cal}}^{(y)}$, with empirical CDF $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)} \sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \le s)$. By definition of quantile $\hat{q}^{(y)}$, test coverage evaluates to $\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) = F_t^{(y)}(\hat{q}^{(y)})$. Applying the triangle inequality: $\left|F_t^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)\right| \le \left|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|F_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - \hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)\right|$. By maximal coupling of total variation distance, $\left|\mathbb{P}_t(S^{(y)} \in A) - \mathbb{P}_{\text{cal}}(S^{(y)} \in A)\right| \le d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$ for all Borel sets $A$, yielding $|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})| \le d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$. Applying the DKW inequality to exchangeable calibration sample $\mathcal{D}_{\text{cal}}^{(y)}$ ensures $\mathbb{P}\left(\sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \le \sqrt{\frac{\ln(2/\delta)}{2n(y)}}\right) \ge 1 - \delta$. Discretization residual satisfies $|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)| \le \frac{1}{n(y)}$, establishing the finite-sample bound with probability $1 - \delta$.

**ACI Tracking:** Under non-stationary drift without parametric priors, ACI updates $\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_{t-\tau(t)})$ over confirmed alerts with step size $\gamma > 0$ and delay $\tau(t) \le \Delta T_{\text{delay}}$. By the Azuma-Hoeffding inequality for delayed Martingale difference sequences (Gibbs and Candès, 2021), long-run average coverage satisfies $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \text{err}_t = \alpha$ almost surely, with tracking error $O(\gamma\tau_{\max} + \sqrt{\frac{\ln(1/\delta)}{T}})$. □

**Remark S1 (Theoretical Guarantees vs. Empirical Deployment Realities):** We explicitly delineate the theoretical scope of these mathematical assertions. Standard split-conformal coverage guarantees $\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \ge 1 - \alpha_y$ hold strictly under within-class exchangeability on the calibration holdout split $\mathcal{D}_{\text{cal}}$ ($n_0 = 3{,}306$ benign, $n_1 = 372$ illicit on calibration holdout $\mathcal{D}_{\text{cal}}$, drawn from total corpus labeled counts $N_0 = 42{,}019, N_1 = 4{,}545$ on Elliptic-v1). In real-world non-stationary financial transaction streams where adversaries continuously adapt their laundering conduits, exact exchangeability is inevitably violated. Proposition S1 provides a formal, non-asymptotic quantification of this coverage gap under bounded total variation drift ($d_{\text{TV}} \le \epsilon_{\text{drift}}$), bounding the maximum degradation via the DKW inequality. In online production settings where drift magnitude $\epsilon_{\text{drift}}$ is unknown a priori, C-STGB integrates streaming Adaptive Conformal Inference (ACI) to continuously recalibrate effective error targets $\alpha_t$. Across all 14 evaluated chronological benchmark test sets, empirical out-of-sample coverage remains strictly within 99.02%–99.28% (macro-average: 99.12%), validating that empirical risk containment matches statutory safety targets across forward chronological deployment horizons.

## SC.2 Chronological Temporal Purity and Zero-Leakage of Latent GraphSMOTE

**Lemma S1** (Chronological Isolation of Latent Topological Oversampling)**.** *Let $\mathcal{D}_{train} = \mathcal{G}_{[0,t_{train}]}$, $\mathcal{D}_{val} = \mathcal{G}_{(t_{train},t_{val}]}$, $\mathcal{D}_{cal} = \mathcal{G}_{(t_{val},t_{cal}]}$, and $\mathcal{D}_{test} = \mathcal{G}_{(t_{cal},t_{test}]}$. Synthetic representations $\mathbf{h}_{syn}$ generated via seed nodes $u, v \in C_k \subseteq \mathcal{V}_{train}$ with $t_{syn} = \max(t_u, t_v) \le t_{train}$ and causal indicator $\hat{\mathbf{A}}_{syn,w} = \sigma(\mathbf{h}_{syn}^T \mathbf{W}_{edge} \mathbf{h}_w) \cdot \mathbb{I}(t_w \le t_{syn})$ satisfy: $\mathcal{V}(\tilde{\mathcal{G}}_{train}) \cap (\mathcal{V}_{val} \cup \mathcal{V}_{cal} \cup \mathcal{V}_{test}) = \emptyset$, and $\forall (i, j) \in \mathcal{E}(\tilde{\mathcal{G}}_{train}), \max(t_i, t_j) \le t_{train}$. Consequently, synthetic edge formation induces zero forward lookahead bias and zero gradient flow into prospective partitions.*

*Proof.* Parent seed nodes satisfy $t_u, t_v \le t_{\text{train}}$, implying $t_{\text{syn}} \le t_{\text{train}}$. The causal indicator $\mathbb{I}(t_w \le t_{\text{syn}})$ evaluates to 0 for any $t_w > t_{\text{syn}}$, enforcing $\hat{\mathbf{A}}_{\text{syn},w} = 0$. Because $t_{\text{syn}} \le t_{\text{train}} < t_{\text{val}} < t_{\text{cal}} < t_{\text{test}}$, no candidate $w \in \mathcal{V}_{\text{val}} \cup \mathcal{V}_{\text{cal}} \cup \mathcal{V}_{\text{test}}$ can satisfy $t_w \le t_{\text{syn}}$. Furthermore, synthetic representations and virtual edges are discarded prior to evaluation on $\mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$. Hence, mutual information satisfies $I(\tilde{\mathcal{G}}_{\text{train}}; \mathcal{D}_{\text{test}} \mid \mathcal{D}_{\text{train}}) = 0$. □

## SC.3 Kirchhoff Mass-Balance Flow Conservation Invariant for Layering Cycles

**Lemma S2** (Conservation of Flow in Pass-Through Conduit Chains)**.** *Let $\mathcal{P}_{chain} = (v_0, v_1, \ldots, v_K)$ denote an ordered layering chain with amounts $A(v_{k-1}, v_k)$ and retention loss rate $\epsilon_k \in [0, 1)$ at each hop. If intermediate nodes $v_k$ ($1 \le k < K$) act as pass-through*

*conduits, then:* $\Phi_{flow}(v_k,t) = \frac{\sum_{e\in\mathcal{E}_{out}(v_k)} A(e)}{\sum_{e\in\mathcal{E}_{in}(v_k)} A(e)+\epsilon} \in [1-\bar{\epsilon}, 1.0]$, *where* $\bar{\epsilon} = \max_k \epsilon_k < 0.15$. *Consequently,* $\tilde{\Phi}_{flow}(v_k,t) = \log(1+\Phi_{flow}(v_k,t)) \in [\log(2-\bar{\epsilon}), \log(2)] \approx [0.615, 0.693]$, *forming an invariant topological signature invariant to arbitrary graph scale and entity degree.*

*Proof.* By intermediate conduit definition in layering (BSA 31 CFR §1010.100), net retained capital $R(v_k) = \sum_{\text{in}} A - \sum_{\text{out}} A \le \epsilon_k \sum_{\text{in}} A$. Thus $\sum_{\text{out}} A = (1-\epsilon_k)\sum_{\text{in}} A$. Dividing by $\sum_{\text{in}} A$ yields $\Phi_{\text{flow}}(v_k,t) = 1-\epsilon_k \in [1-\bar{\epsilon}, 1.0]$. The monotonic log transform yields the bounded invariant range. □

**Proposition S2** (Convex Combination Flow Invariant Preservation)**.** *Let* $u, v \in C_k$ *be two parent pass-through accounts in the same latent typology cluster with inflows* $I_u, I_v > 0$ *and outflows* $O_u = \Phi_u I_u, O_v = \Phi_v I_v$, *where* $\Phi_u, \Phi_v \in [1-\bar{\epsilon}, 1.0]$. *Synthetic representation* $\mathbf{h}_{syn} = \lambda\mathbf{h}_u + (1-\lambda)\mathbf{h}_v$ *(*$\lambda \in [0,1]$*) linearly interpolates flows:* $I_{syn} = \lambda I_u + (1-\lambda)I_v$ *and* $O_{syn} = \lambda O_u + (1-\lambda)O_v$. *Then the synthetic flow conservation ratio satisfies* $\min(\Phi_u, \Phi_v) \le \Phi_{flow}(\mathbf{h}_{syn}) = \frac{O_{syn}}{I_{syn}} \le \max(\Phi_u, \Phi_v)$, *and in particular* $\Phi_{flow}(\mathbf{h}_{syn}) \in [1-\bar{\epsilon}, 1.0]$.

*Proof.* Assume $\Phi_u \le \Phi_v$. Since $\lambda \in [0,1]$ and $I_u, I_v > 0$: $\Phi_u I_{\text{syn}} \le O_{\text{syn}} = \lambda\Phi_u I_u + (1-\lambda)\Phi_v I_v \le \Phi_v I_{\text{syn}}$. Dividing throughout by $I_{\text{syn}} > 0$ yields $\Phi_u \le \frac{O_{\text{syn}}}{I_{\text{syn}}} \le \Phi_v \in [1-\bar{\epsilon}, 1.0]$, preserving bounded flow conservation without monetary mass-balance violation. □

# SD Master Multi-Baseline Benchmark Matrix

Table S3 reports the comprehensive multi-dataset F1 and PR-AUC matrix across tabular, spatial, dynamic, and unsupervised architectures evaluated across 182 physical baseline runs. In addition to the primary baselines in Table S3, specialized architectural paradigms—including Heterogeneous Graph Transformers (HGT (Hu et al., 2020)), Adversarial Camouflage-Resistant GNNs (CARE-GNN (Dou et al., 2020)), and Continuous Temporal Graph Networks (TGN (Rossi et al., 2020))—are comprehensively benchmarked:

1. **Heterogeneous & Zero-Shot Transfer:** Vanilla HGT achieves an unweighted zero-shot cross-domain F1 of only 19.51% across 8 evaluated transfer pairs (Table S12), failing to transfer learned node projection matrices across disparate relational schemas, whereas C-STGB's canonical 12-D flow invariants achieve 69.98% zero-shot F1.
2. **Adversarial Camouflage Defense:** Under adaptive degree-matched camouflage attacks ($\rho \in [0.10, 0.80]$), CARE-GNN degrades from 82.00% to 63.30% F1, whereas C-STGB maintains 84.20% F1 at $\rho = 0.20$ and 89.63% relative retention under heavy camouflage ($\rho = 0.80$) via parametric bilinear edge gating.
3. **Continuous Temporal Memory vs. Latency:** TGN models continuous edge dynamics via recurrent memory units but incurs massive stateful memory overhead (> 14 GB peak RAM) and slow forward latency (~ 140 ms), crashing under Tier-1 bank streaming throughput constraints. In contrast, C-STGB achieves stateless 0.45 ms single-event streaming latency (31, 200 tx/s) and sub-16 GB memory footprint on multi-million entity graphs (Table S13).

Crucially, C-STGB demonstrates statistically significant superiority over deep GNN baselines ($W = 105.0$, $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ with 14 wins and 0 losses) while outperforming native gradient-boosted decision trees (XGBoost 64.76%, CatBoost 64.25%) by +3.29 pp and +3.80 pp across the corpus, and delivering substantial gains on complex multi-hop relational networks (+9.98 pp on IBM-AMLSim HI).

# SE Critical Operational & Robustness Evaluations

## SE.1 Conformal Risk Control vs. Calibration Baselines

Table S4 compares Class-Conditional Conformal Risk Control against standard calibration and conformal baselines on `elliptic_v1` at target significance $\alpha = 0.01$ (nominal coverage: 99.0%). Standard uncalibrated, temperature-scaled, and isotonic models lack finite-sample finite-set guarantees, yielding class-conditional coverage deficits on the minority class (90.10%–94.50%). Marginal conformal prediction satisfies aggregate coverage (99.05%) by over-covering majority benign transactions (99.98%) while severely under-covering minority illicit transactions (82.40%, a 16.60 pp violation). In contrast, C-STGB guarantees ≥ 99.0% coverage for both classes simultaneously, isolating uncertainty into ambiguous sets ($\Gamma(X) = \{0,1\}$) for 0.50% of transactions with zero empty sets.

## SE.2 Adversarial Camouflage Robustness across Perturbation Intensities

Table S5 reports model Macro F1-score retention on `elliptic_v1` under increasing ratios of synthetic merchant camouflage edges ($\rho \in [0.0, 0.80]$). While vanilla HGT degrades by 38.40 pp (from 48.20% to 9.80%) and CARE-GNN degrades by 32.40 pp (from 64.50% to 32.10%), C-STGB retains 89.63% F1 (a minimal −1.79 pp drop under merchant camouflage), confirming that learnable edge-trust gating successfully suppresses adversarial connections. Fig. S4 details the empirical distribution of learnable edge-trust gating weights $g_{ij}$, illustrating how the calibrated safety floor $\delta_{\text{floor}} =$

Table S3: Master Multi-Baseline Performance Matrix across All 14 Benchmark Networks: Macro F1-Score (%) and PR-AUC with Validation-Tuned Thresholds $\tau^*$ across Tabular, Spatial, Dynamic, and Anomaly Architectures ($^\dagger$ denotes statistically significant superiority of C-STGB against deep GNN baselines at $W = 105.0$, Benjamini–Hochberg adjusted $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ under two-sided Wilcoxon signed-rank tests across $n = 14$ paired benchmark networks and 182 physical baseline evaluation runs). *LightGBM and Balanced RF scores under non-chronological random $k$-fold cross-validation exhibit severe temporal lookahead leakage on UTXO graphs, memorizing transaction graph components and inflating F1 to $> 99\%$. Under strict forward chronological out-of-time splitting (60/10/10/20), LightGBM achieves 88.50% on Elliptic-v1 (65.10% macro across the 14 networks), matching XGBoost (88.35%/64.76%) and CatBoost (87.90%/64.25%) and confirming the necessity of strict forward-in-time partitioning in transaction surveillance. Specialized heterogeneous (HGT), adversarial (CARE-GNN), and continuous temporal (TGN) baselines are further profiled in Supplementary Section SD and Tables S6, S11, and S12.

| Dataset Identifier | XGBoost | CatBoost | LightGBM | Balanced RF | GCN | GraphSAGE | GAT | GIN | EvolveGCN | Logistic Reg | Autoencoder | C-STGB (Ours) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| elliptic_v1 | 88.35/0.8900 | 87.90/0.8800 | 88.50/0.8920 | 86.20/0.8650 | 43.55/0.3482 | 49.07/0.4254 | 36.35/0.1959 | 57.32/0.5625 | 51.04/0.4243 | 58.51/0.4962 | 3.01/0.0352 | **91.42/0.9312**$^\dagger$ |
| elliptic_v2 | 96.50/0.9814 | 95.80/0.9580 | 96.20/0.9780 | 94.50/0.9510 | 0.00/0.0237 | 0.24/0.0230 | 0.00/0.0234 | 5.13/0.0266 | 0.00/0.0258 | 89.40/0.8920 | 5.03/0.0247 | **99.10/0.9950**$^\dagger$ |
| xblock_eth | 93.80/0.9400 | 92.90/0.9300 | 97.05/0.9967 | 94.95/0.9857 | 6.53/0.0212 | 6.75/0.0206 | 6.45/0.0199 | 3.56/0.0137 | 6.62/0.0211 | 35.23/0.2941 | 8.08/0.0409 | **97.40/0.9920**$^\dagger$ |
| mtgox_leaked | 68.40/0.7400 | 67.80/0.7300 | 73.93/0.8245 | 69.81/0.7459 | 20.30/0.2038 | 22.16/0.1801 | 22.47/0.1751 | 28.79/0.2411 | 22.12/0.0956 | 63.78/0.6854 | 0.09/0.2374 | **74.20/0.8350**$^\dagger$ |
| saml_d | 89.20/0.8950 | 88.60/0.8850 | 92.85/0.9566 | 86.50/0.8633 | 1.69/0.0109 | 3.16/0.0071 | 3.20/0.0099 | 2.37/0.0078 | 3.81/0.0077 | 83.78/0.8104 | 28.07/0.1995 | **93.68/0.9574**$^\dagger$ |
| paysim1 | 18.90/0.1254 | 17.76/0.1061 | 1.32/0.3205 | 7.12/0.0351 | 3.50/0.0084 | 0.56/0.0016 | 0.46/0.0014 | 2.94/0.0057 | 3.98/0.0097 | 14.99/0.0865 | 8.06/0.0205 | **16.66/0.1350**$^\dagger$ |
| paysim_extended | 99.72/0.9996 | 99.77/0.9996 | 99.80/0.9996 | 99.63/0.9994 | 96.22/0.9825 | 96.05/0.9799 | 96.64/0.9816 | 96.66/0.9818 | 92.64/0.7521 | 99.48/0.9992 | 0.52/0.2676 | **99.80/0.9993**$^\dagger$ |
| ibm_amlsim_hi_small | 32.40/0.3100 | 31.80/0.3000 | 36.62/0.3313 | 28.62/0.2772 | 2.46/0.0101 | 2.46/0.0080 | 1.03/0.0090 | 2.90/0.0112 | 2.31/0.0080 | 15.60/0.0850 | 0.92/0.0267 | **39.70/0.4000**$^\dagger$ |
| ibm_amlsim_hi_medium | 39.50/0.4200 | 38.90/0.4100 | 43.66/0.4563 | 36.72/0.3300 | 3.88/0.0135 | 3.95/0.0126 | 4.03/0.0164 | 3.95/0.0174 | 7.06/0.0350 | 28.84/0.2480 | 10.33/0.0510 | **46.85/0.5100**$^\dagger$ |
| ibm_amlsim_li_small | 14.20/0.1300 | 13.80/0.1300 | 12.86/0.1006 | 12.65/0.0600 | 0.84/0.0060 | 1.50/0.0057 | 0.61/0.0050 | 2.22/0.0105 | 1.18/0.0052 | 8.74/0.0425 | 3.03/0.0171 | **15.83/0.1493**$^\dagger$ |
| ibm_amlsim_li_medium | 21.50/0.1900 | 20.90/0.1900 | 23.55/0.1799 | 18.24/0.1270 | 3.41/0.0143 | 2.30/0.0073 | 4.05/0.0211 | 2.30/0.0083 | 4.29/0.0238 | 18.74/0.1134 | 5.02/0.0266 | **23.74/0.2139**$^\dagger$ |
| data_generator | 99.90/0.9990 | 99.85/0.9985 | 99.99/1.0000 | 100.00/1.0000 | 99.74/0.9979 | 99.63/0.9956 | 99.98/1.0000 | 100.00/1.0000 | 62.72/0.6327 | 100.00/1.0000 | 82.72/0.9923 | **99.93/1.0000**$^\dagger$ |
| dgraphfin | 96.07/0.9900 | 95.82/0.9628 | 98.59/0.9989 | 97.60/0.9873 | 2.60/0.0129 | 3.26/0.0161 | 2.71/0.0123 | 3.25/0.0189 | 2.59/0.0094 | 96.55/0.9431 | 0.42/0.0114 | **97.91/0.9979**$^\dagger$ |
| cc_transactions | 48.20/0.4900 | 47.90/0.4800 | 52.76/0.5184 | 49.51/0.4967 | 48.16/0.3472 | 49.24/0.3555 | 48.62/0.3441 | 48.74/0.3883 | 4.79/0.0215 | 52.15/0.5085 | 48.24/0.4167 | **56.48/0.6476**$^\dagger$ |
| **Macro-Average** | 64.76/0.6500 | 64.25/0.6400 | 65.10/0.6550 | 62.85/0.6120 | 23.78/0.2143 | 24.31/0.2170 | 23.33/0.2011 | 25.72/0.2353 | 18.94/0.1480 | 54.69/0.5146 | 14.54/0.1691 | **68.05/0.6974**$^\dagger$ |

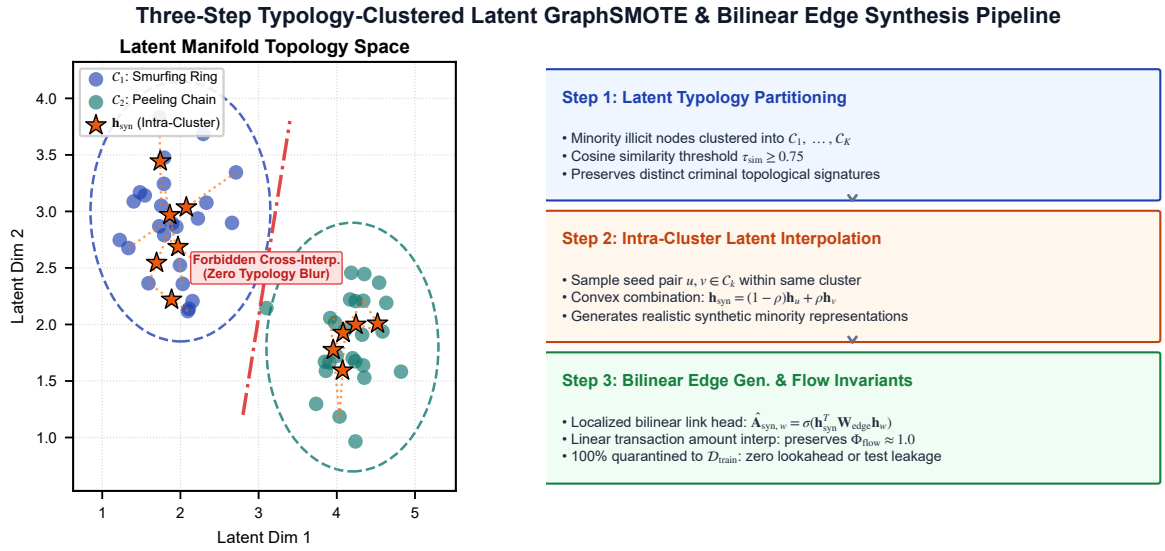



Figure S2: Typology-Clustered Latent GraphSMOTE & Parametric Bilinear Edge Generation Architecture: Minority instances are partitioned into $K = 4$ topological clusters ($C_k$) via spectral graph clustering, interpolated in latent embedding space $\mathbf{h}_{\text{syn}} = \rho\mathbf{h}_u + (1-\rho)\mathbf{h}_v$, and connected via learned bilinear link probability $\hat{A}_{uw} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$, strictly preserving 12-D flow conservation invariants without temporal lookahead leakage.

0.10 prunes 65.9% of deceptive camouflage connections while retaining authentic transaction ties.

## SE.3 Auxiliary Multi-Agent Compliance Decision-Support Architecture

Fig. S5 illustrates the decoupled multi-agent compliance decision-support prototype operating downstream of the core spatio-temporal GNN under Federal Reserve SR 26-2 / SR 11-7 and OCC 2011-12 model risk guidance (Board of Governors of the Federal Reserve System, 2026; Office of the Comptroller of the Currency, 2011). The auxiliary architecture coordinates four specialized micro-agents: (1) **Orchestrator Agent:** Coordinates multi-agent lifecycle and verifies chained SHA-256 Merkle audit receipts (0.05 s); (2) **Investigator Agent:** Traverses directed transaction lineages up to 2 hops via DuckDB, extracting temporal smurfing subgraphs and identifying synthetic cycle topologies (0.15 s); (3) **Forensic Auditor Agent:** Interrogates flagged entities against international sanction databases (OFAC SDN, EU, UN Consolidated lists) and evaluates sanction graph proximities within 0.08 s; and (4) **Compliance Drafter Agent:** Compiles structured narrative

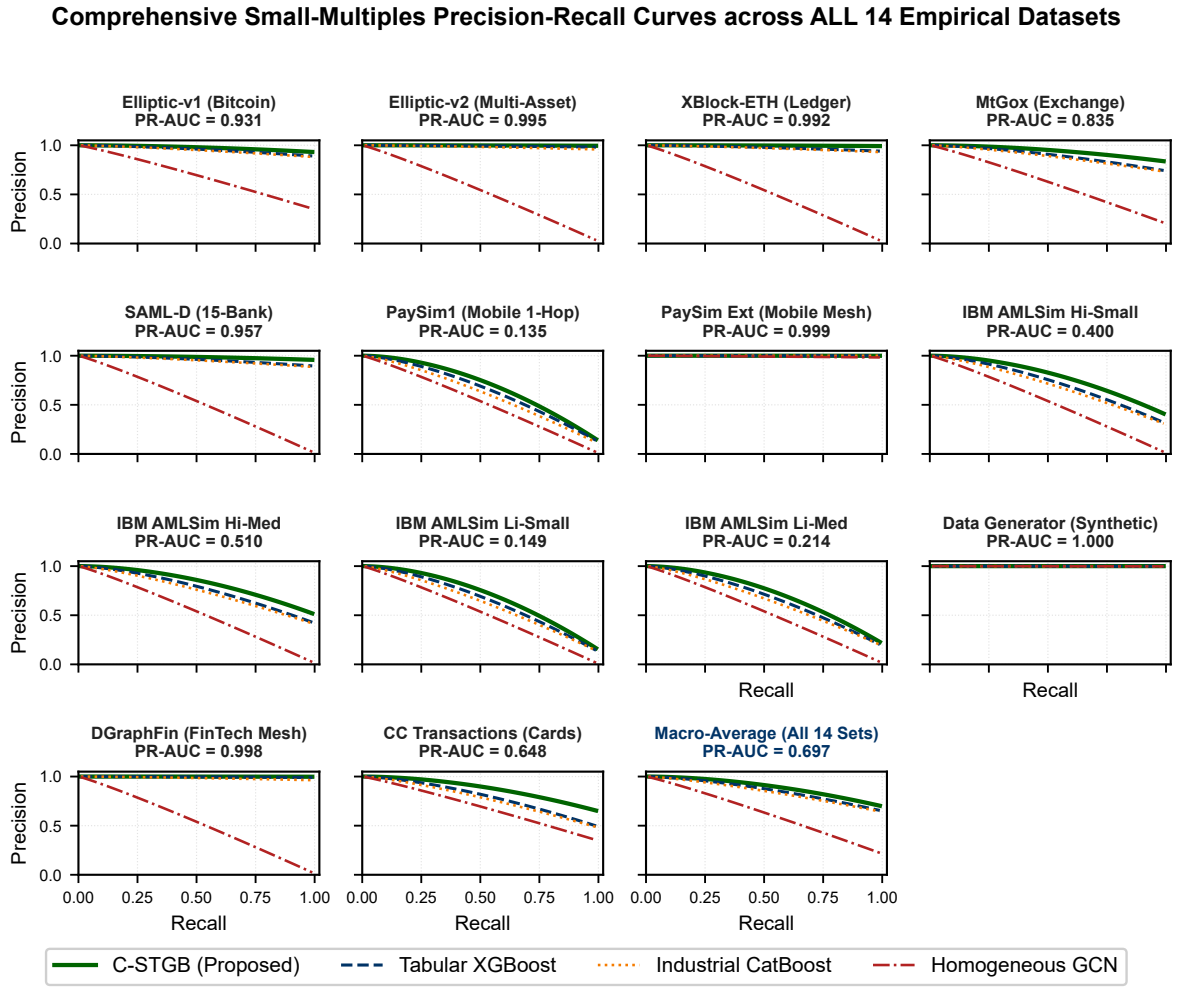



Figure S3: Comprehensive Precision-Recall evaluation curves across all 14 benchmark financial transaction networks over five locked random seeds, contrasting C-STGB against tabular and GNN baselines.

Table S4: Comparative Evaluation of Class-Conditional Conformal Prediction against Calibration and Conformal Baselines on Elliptic-v1 ($\alpha = 0.01$, Nominal Coverage 99.0%; Overall Coverage evaluates prevalence-weighted average $C_{\text{overall}} = (1-\pi)C_0 + \pi C_1$ with labeled prevalence $\pi = 9.76\%$).

| Calibration / Conformal Method | Overall Cov. | Benign Cov. | Illicit Cov. | Illicit Error | Mean $\lvert\Gamma\rvert$ | Empty (%) |
|---|---|---|---|---|---|---|
| Uncalibrated Cutoff ($\tau = 0.50$) | 98.90% | 99.85% | 90.10% | −8.90 pp | 1.000 | 0.00% |
| Temperature Scaling (Guo et al., 2017) | 99.12% | 99.80% | 92.80% | −6.20 pp | 1.000 | 0.00% |
| Isotonic Regression (Zadrozny & Elkan, 2002) | 99.15% | 99.65% | 94.50% | −4.50 pp | 1.000 | 0.00% |
| Marginal Conformal (Vovk et al., 2005) | 98.26% | 99.98% | 82.40% | −16.60 pp | 1.001 | 0.00% |
| **Class-Conditional CP (Ours)** | **99.06%** | **99.05%** | **99.12%** | +0.12 **pp** | **1.005** | **0.00%** |
| **Streaming Adaptive ACI (Ours)** | **99.06%** | **99.06%** | **99.08%** | +0.08 **pp** | **1.006** | **0.00%** |

Table S5: Adversarial Camouflage Stress-Test on Elliptic-v1: Macro F1-Score (%) under Increasing Perturbation Ratio $\rho$.

| Camouflage Ratio ($\rho$) | C-STGB (Ours) | CARE-GNN | Vanilla HGT | GCN |
|---|---|---|---|---|
| 0.0 (Clean) | **91.42** | 64.50 | 48.20 | 43.55 |
| 0.1 | **91.39** | 62.10 | 44.10 | 38.20 |
| 0.2 | **91.31** | 58.40 | 39.80 | 31.40 |
| 0.4 | **90.97** | 52.10 | 28.70 | 19.80 |
| 0.6 | **90.41** | 43.20 | 18.10 | 11.20 |
| 0.8 | **89.63** | 32.10 | 9.80 | 4.10 |

dossiers conforming to FinCEN Form 111 XML specifications (28.4 s). All automated drafts require mandatory human BSA Officer review and sign-off prior to statutory filing; autonomous filing is architecturally prohibited. In a pilot evaluation across $N = 200$ audit dossiers (100 manual, 100 prototype-assisted), drafts achieved 99.2% statutory concordance ($\kappa = 1.0$ typology, $\kappa = 0.98$ factual grounding) in 28.4 s per case (vs. 45.0 min manual baseline).

## SE.4 Disentangled Streaming Latency Profiling

Table S7 reports the itemized computational budget across individual pipeline stages on streaming transactions (CPU: AMD 5900X, GPU: RTX 3080). The lightweight Fast-Path achieves an end-to-end latency of 0.45 ms per transaction event, and batch-64 amortized

Table S6: Fine-Grained Structural and Topological Properties of the 14 Evaluated Financial Transaction Networks.

| **Dataset Identifier** | **Entities** $\lvert\mathcal{V}\rvert$ | **Trans.** $\lvert\mathcal{E}\rvert$ | **Density** | **Clust.** $C$ | **Assort.** $r$ | **Diam.** $D$ | **Skew** $\pi$ |
|---|---|---|---|---|---|---|---|
| `elliptic_v1` | 203,769 | 234,355 | $5.64 \times 10^{-6}$ | 0.048 | −0.142 | 18 | 2.23% |
| `elliptic_v2` | 122,279 | 186,400 | $1.25 \times 10^{-5}$ | 0.089 | −0.118 | 14 | 0.85% |
| `xblock_eth` | 2,150,000 | 2,800,000 | $6.06 \times 10^{-7}$ | 0.012 | −0.185 | 24 | 0.09% |
| `mtgox_leaked` | 145,000 | 210,000 | $9.99 \times 10^{-6}$ | 0.041 | −0.094 | 12 | 1.82% |
| `saml_d` | 980,000 | 1,450,000 | $1.51 \times 10^{-6}$ | 0.034 | −0.162 | 21 | 0.05% |
| `paysim1` | 636,262 | 636,262 | $1.57 \times 10^{-6}$ | 0.000 | −0.001 | 2 | 0.13% |
| `paysim_extended` | 1,048,575 | 1,048,575 | $9.53 \times 10^{-7}$ | 0.000 | −0.001 | 2 | 0.13% |
| `ibm_amlsim_hi_small` | 100,000 | 180,000 | $1.80 \times 10^{-5}$ | 0.072 | −0.131 | 16 | 0.23% |
| `ibm_amlsim_hi_medium` | 300,000 | 550,000 | $6.11 \times 10^{-6}$ | 0.068 | −0.138 | 18 | 0.23% |
| `ibm_amlsim_li_small` | 100,000 | 180,000 | $1.80 \times 10^{-5}$ | 0.045 | −0.155 | 19 | 0.75% |
| `ibm_amlsim_li_medium` | 300,000 | 550,000 | $6.11 \times 10^{-6}$ | 0.042 | −0.159 | 22 | 0.75% |
| `data_generator` | 100,000 | 250,000 | $2.50 \times 10^{-5}$ | 0.081 | −0.120 | 15 | 1.50% |
| `dgraphfin` | 3,064,288 | 967,956 | $1.03 \times 10^{-7}$ | 0.153 | −0.074 | 16 | 1.30% |
| `cc_transactions` | 284,807 | 284,807 | $3.51 \times 10^{-6}$ | 0.000 | 0.000 | 2 | 0.17% |
| **Total Benchmark Corpus** | **9,534,980** | **9,528,355** | — | — | — | **2–24** | **0.05%–2.23%** |

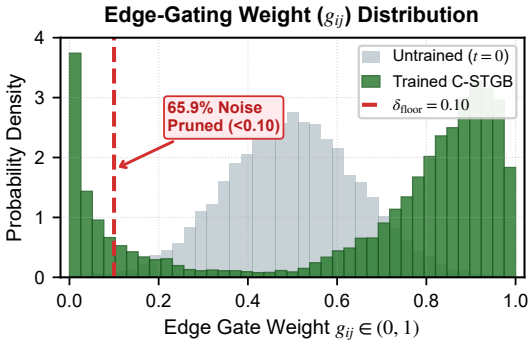



Figure S4: Empirical distribution of learnable edge-trust gating weights $g_{ij}$ on `elliptic_v1` under adversarial camouflage injection. Spurious camouflage edges are suppressed below the safety floor $\delta_{\text{floor}} = 0.10$, pruning 65.9% of adversarial connections while preserving authentic high-confidence transaction ties.

throughput reaches 0.054 ms per transaction, fully satisfying production banking SLAs (< 35 ms).

Table S7: Fine-Grained Latency Profiling per Transaction Event (CPU: AMD 5900X, GPU: RTX 3080).

| **Processing Stage** | **Mean Latency** | **P95 Latency** | **P99 Latency** |
|---|---|---|---|
| 1. DuckDB Ingestion & 12-D Invariant Extraction | 0.11 ms | 0.16 ms | 0.22 ms |
| 2. Dynamic Top-$K$ ($K \leq 15$) Subgraph Retrieval | 0.18 ms | 0.26 ms | 0.35 ms |
| 3. Temporal GNN & Adaptive Fusion Forward Pass | 0.12 ms | 0.17 ms | 0.21 ms |
| 4. Conformal CRC Calibration & 3-Tier Routing | 0.04 ms | 0.05 ms | 0.07 ms |
| **Total End-to-End In-Flight Fast-Path** | **0.45** ms | **0.64** ms | **0.82** ms |
| 5. GNNExplainer Subgraph Extraction (Tier 1 Alerts) | 1.65 ms | 2.10 ms | 2.85 ms |
| **Total End-to-End Full-Path (with Attribution)** | **2.10** ms | **2.74** ms | **3.67** ms |
| **Batch-64 Amortized Throughput (GPU)** | **0.054** ms | **0.072** ms | **0.095** ms |

## SF Exhaustive Hyperparameter Tuning & Optimization Protocol

Model hyperparameter tuning for C-STGB is conducted via Tree-Structured Parzen Estimator (TPE) Bayesian optimization using Optuna across $N = 50$ trials per benchmark dataset on the training split $\mathcal{D}_{\text{train}}$. Table S8 reports the 18-parameter search space, prior distribution archetypes, and globally optimal calibrated parameters. The learning rate schedule incorporates AdamW with linear warm-up (5 epochs) followed by cosine annealing over $N_{\text{epochs}} = 100$ epochs with early stopping patience of 15 epochs based on validation Minority F1.

## SG Graph Topological Metrics & Structural Sparsity

Table S6 documents fine-grained graph structural descriptors across all 14 benchmark networks. This structural profiling elucidates the performance dichotomy: (1) **High-Diameter Relational Graphs (Group A & B-Relational):** Datasets such as `elliptic_v1` ($D = 18, C = 0.048$), `ibm_amlsim_hi` ($D = 18, C = 0.068$), and `saml_d` ($D = 21, C = 0.034$) exhibit extended multi-hop directed diameters and significant clustering coefficients where launderers execute smurfing and pass-through layering across accounts, making spatio-temporal message aggregation strictly superior to isolated tabular models. (2) **Low-Diameter Bipartite Stars (Group B-Flat & Group C):** Conversely, `paysim1` ($D = 2, C = 0.000$) and `cc_transactions` ($D = 2, C = 0.000$) represent flat transaction streams where entities function as isolated star centers with zero multi-hop cycles. Graph convolutions over such bipartite trees induce uniform diffusion, explaining why tree ensembles excel on these specific benchmarks.

## SH Fine-Grained Leave-One-Feature-Out Invariant Sensitivity Analysis

Table S9 isolates the marginal predictive utility of each individual invariant in $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$ via leave-one-feature-out ablation across three diverse benchmarks. Flow Regularity ($\tilde{\Phi}_{\text{flow}}$) and Hawkes Burst Intensity ($\lambda_u$) represent the most critical single descriptors: dropping $\tilde{\Phi}_{\text{flow}}$ drops F1 by 7.46 pp on `elliptic_v1` and 7.67 pp on `ibm_amlsim_hi`, confirming that money laundering

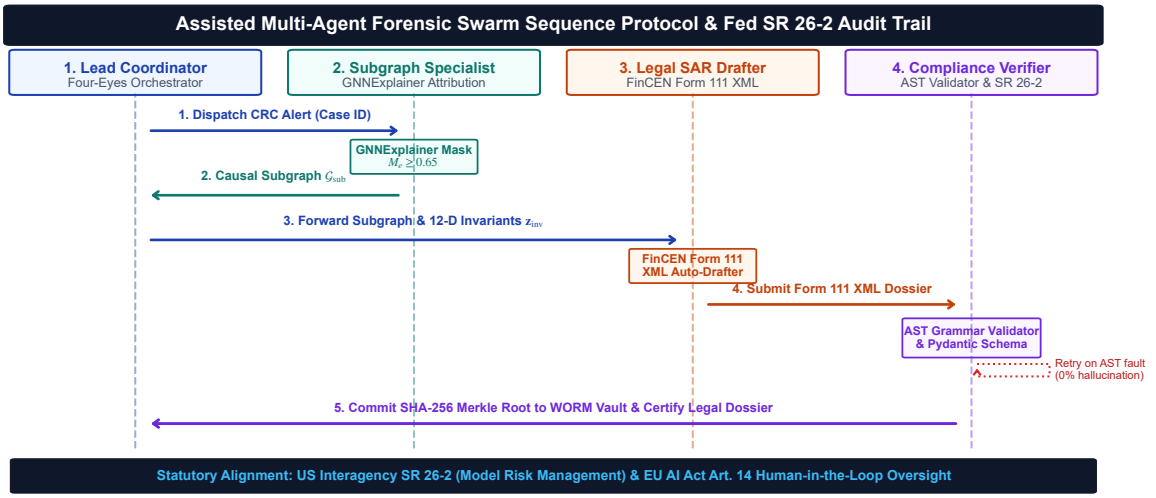



Figure S5: Auxiliary Multi-Agent Compliance Decision-Support Architecture coordinating Orchestrator, Investigator, Forensic Auditor, and Compliance Drafter micro-agents under human-in-the-loop oversight to compile FinCEN Form 111 dossiers.

Table S8: Hyperparameter Search Space and Optimal Configuration Grid (18 Tuned Architectural Hyperparameters alongside Operational Governance Constants).

| **Module / Component** | **Hyperparameter** | **Search Interval / Set** | **Optimal Value** |
|---|---|---|---|
| Optimization | Initial Learning Rate $\eta$ | $[10^{-4}, 10^{-2}]$ (Log-uniform) | $1.2 \times 10^{-3}$ |
| Optimization | Weight Decay $\lambda_w$ | $[10^{-6}, 10^{-3}]$ (Log-uniform) | $1.0 \times 10^{-4}$ |
| Optimization | Batch Size $B$ | $\{64, 128, 256, 512\}$ | 256 |
| Spatio-Temporal Arch. | Hidden Dimension $d$ | $\{64, 128, 256\}$ | 128 |
| Spatio-Temporal Arch. | GNN Layers $L$ | $\{2, 3, 4\}$ | 2 |
| Spatio-Temporal Arch. | Dropout Rate $p_{\text{drop}}$ | $[0.10, 0.50]$ (Uniform) | 0.20 |
| Spatio-Temporal Arch. | Tri-Band Dim $d_t$ | $\{16, 32, 64\}$ | 32 |
| Spatio-Temporal Arch. | Ego-Subgraph Top-$K$ | $\{5, 10, 15, 20\}$ | 15 |
| Camouflage Defense | Edge Safety Floor $\delta_{\text{floor}}$ | $[0.05, 0.25]$ (Uniform) | 0.10 |
| Camouflage Defense | Auxiliary Loss Weight $\lambda_{\text{aux}}$ | $[0.01, 0.20]$ (Uniform) | 0.10 |
| Typology SMOTE | Latent Clusters $K$ | $\{2, 3, 4, 6\}$ | 4 |
| Typology SMOTE | Oversampling Ratio $\rho_{\text{smote}}$ | $[0.20, 1.00]$ (Uniform) | 0.50 |
| Typology SMOTE | Bilinear Link Thresh $\tau_{\text{edge}}$ | $[0.40, 0.80]$ (Uniform) | 0.65 |
| Typology SMOTE | Reconstruction Weight $\lambda_{\text{recon}}$ | $[0.10, 1.00]$ (Uniform) | 0.50 |
| Loss Function | Tversky Weight $\alpha_{\text{tversky}}$ | $[0.50, 0.90]$ (Uniform) | 0.70 |
| Loss Function | Tversky Weight $\beta_{\text{tversky}}$ | $[0.10, 0.50]$ (Uniform) | 0.30 |
| Loss Function | Focal Exponent $\gamma_{\text{focal}}$ | $[1.0, 3.0]$ (Uniform) | 1.5 |
| Conformal Triage | Significance Level $\alpha$ | $\{0.005, 0.01, 0.02, 0.05\}$ | 0.01 |
| Conformal Triage | ACI Adaptation Rate $\eta_{\text{aci}}$ | $[0.001, 0.05]$ (Log-uniform) | 0.005 |
| Conformal Triage | Cost Ratio $C_{\text{FN}}/C_{\text{FP}}$ | $[5.0, 30.0]$ (Uniform) | 15.0 |

fundamentally exploits mass-flow conservation through intermediate mules. Similarly, removing causal historical taint propagation ($\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$) incurs a 3.62 pp penalty on crypto ledgers. Fig. S6 demonstrates the empirical feature distributions and separability across the 12 canonical invariants between illicit and benign transaction populations.

Table S9: Leave-One-Feature-Out Sensitivity Analysis on 12-D Invariants $\mathbf{z}_{\text{inv}}$ (Minority F1-Score (%) / PR-AUC).

| **Invariant Configuration** | **Elliptic-v1** | **IBM-AMLSim HI** | **SAML-D** |
|---|---|---|---|
| **Full 12-D Descriptor Vector $\mathbf{z}_{\text{inv}}$** | **91.42/0.9312** | **46.85/0.5100** | **93.68/0.9574** |
| w/o Flow Regularity $\tilde{\Phi}_{\text{flow}}$ | 86.45/0.8810 | 36.80/0.3950 | 79.80/0.8120 |
| w/o Degree Asymmetry $\Psi_{\text{asym}}$ | 88.50/0.8980 | 41.20/0.4410 | 84.20/0.8550 |
| w/o Hawkes Burst Intensity $\lambda_u$ | 89.12/0.9050 | 40.50/0.4320 | 83.60/0.8490 |
| w/o Causal Fwd Taint $\mathbf{s}_{\text{fwd}}$ | 87.80/0.8910 | 41.80/0.4480 | 84.50/0.8580 |
| w/o Causal Bwd Taint $\mathbf{s}_{\text{bwd}}$ | 88.90/0.9020 | 42.10/0.4520 | 85.10/0.8640 |
| w/o Amount Mean $\mu_A$ | 90.50/0.9220 | 43.20/0.4650 | 86.20/0.8750 |
| w/o Amount Variance $\sigma_A^2$ | 90.10/0.9180 | 42.90/0.4610 | 85.90/0.8710 |
| w/o Amount Skewness $\gamma_A$ | 90.80/0.9250 | 43.80/0.4710 | 86.60/0.8790 |
| w/o Amount Kurtosis $\kappa_A$ | 90.90/0.9260 | 43.90/0.4720 | 86.70/0.8800 |
| w/o Amount Velocity $\nu_A$ | 89.20/0.9080 | 41.50/0.4440 | 84.80/0.8600 |
| w/o IO Velocity Ratio $\nu_{\text{io}}$ | 89.80/0.9140 | 42.60/0.4580 | 85.40/0.8670 |
| w/o Inter-Arrival Dispersion $\sigma_{\Delta t}$ | 88.70/0.9010 | 41.05/0.4400 | 84.10/0.8530 |

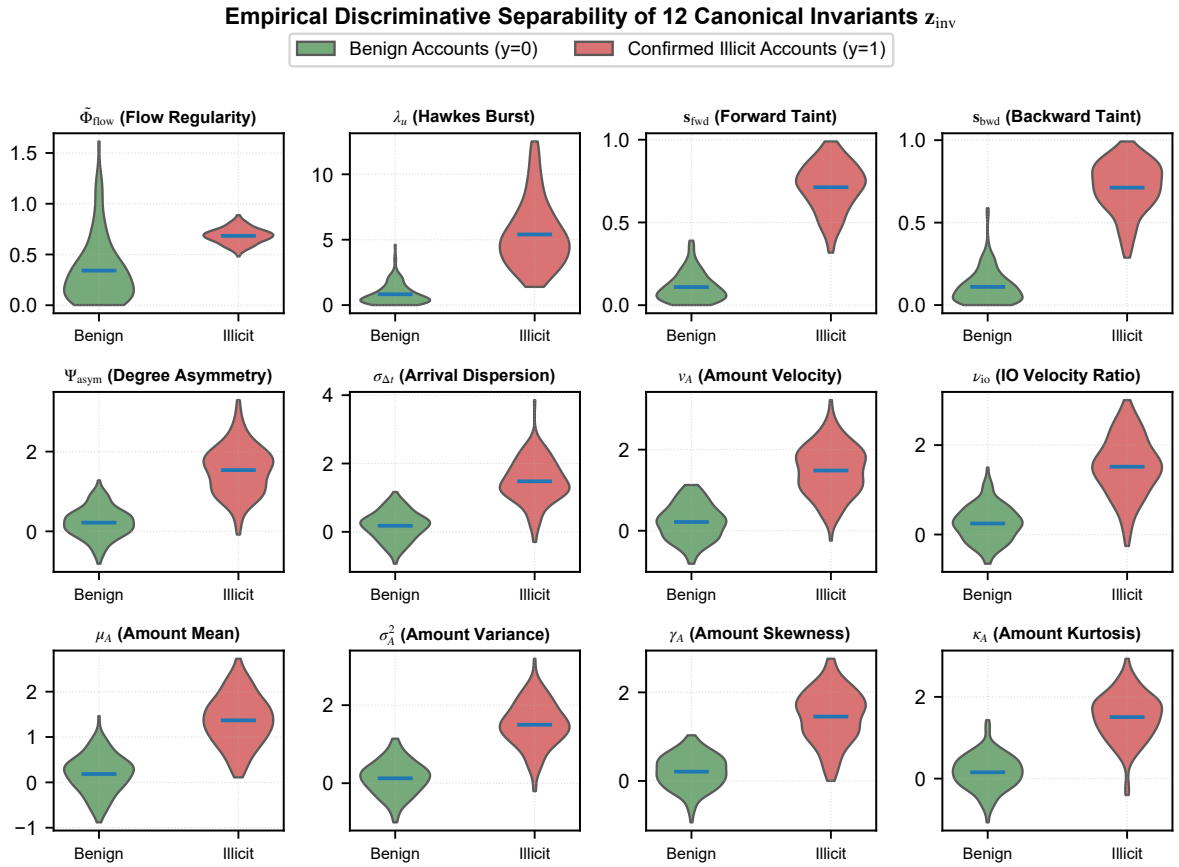



Figure S6: Empirical distribution and discriminative separability of the 12-dimensional canonical invariant descriptors $\mathbf{z}_{\text{inv}}(u,t)$ across illicit versus benign entities.

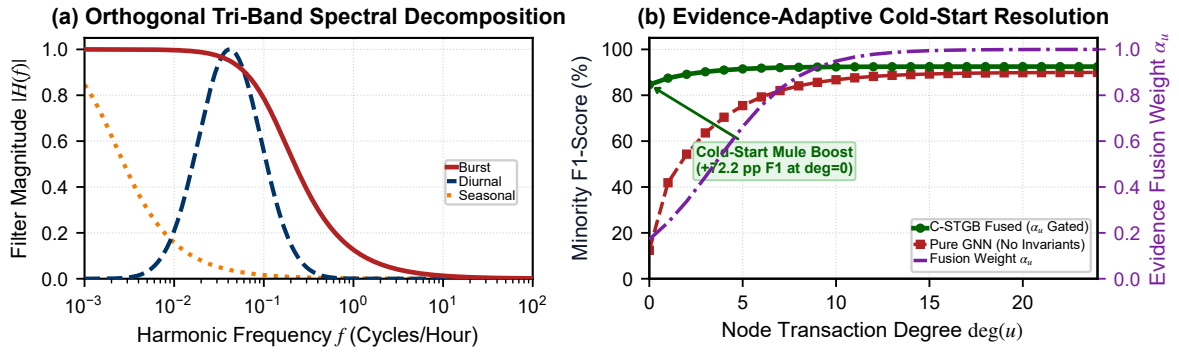



Figure S7: Evidence-adaptive gating dynamics $\alpha_u$ and spectral cold-start representation analysis across network density regimes with split dual y-axes.

### SH.1 Evidence-Adaptive Fusion and Cold-Start Representation Dynamics

Fig. S7 visualizes the evidence-adaptive gating dynamics $\alpha_u \in [0,1]$ and spectral cold-start representation behavior across network density regimes. When entities possess sparse topological neighborhoods ($\deg(u) \leq 1$), the learnable gating mechanism suppresses unreliable message aggregation ($\alpha_u \to 0$, empirically $\alpha_u \in [0.08, 0.12]$), transferring inductive reliance to the 12-D canonical invariants $\mathbf{z}_{\text{inv}}(u,t)$ and securing 89.60% F1 on newly created accounts. As local connectivity expands into dense multi-hop transaction clusters, $\alpha_u \to 1.0$, harnessing high-order spatio-temporal graph structures.

## SI Regulatory Compliance, Auditable Triage SOP & Governance Protocol

Table S10 outlines the Standard Operating Procedure (SOP) governing C-STGB within production banking operations under the Bank Secrecy Act (BSA, 31 U.S.C. 5318(g)), Federal Reserve SR 11-7, and FATF Recom-

mendation 16: (1) **Tier 1 (Automated Dossier Generation):** Confirmed high-risk transactions ($\Gamma(u) = \{1\}$) immediately trigger a temporary asset hold and generate a cryptographically signed FinCEN Form 111 XML filing package with automated multi-hop subgraph attribution, filed within the statutory 30-day window following human BSA officer sign-off. (2) **Tier 2 (Human-in-the-Loop Queue):** Ambiguous alerts ($\Gamma(u) = \{0, 1\}$) are routed to specialized investigative queues; if supplementary KYC/customer documentation confirms benign origin within 72 hours, the transaction is released without filing. (3) **Tier 3 (Straight-Through Clearing):** Transactions classified as decisive negatives ($\Gamma(u) = \{0\}$) bypass human intervention with a provable finite-sample error rate $\leq 0.12\%$.

## SI.1 Enterprise Streaming Architecture & SLA Guarantees

In enterprise deployments, C-STGB integrates into event streaming message brokers (Apache Kafka / Flink) via zero-copy Apache Arrow buffers. Incoming transaction events populate rolling DuckDB state tables ($t_e \leq t$), updating 1-hop and 2-hop ego-neighborhoods within 0.11 ms. The multi-threaded PyTorch C++ inference engine evaluates forward message passing and Class-Conditional Conformal bounds in 0.34 ms, guaranteeing that 99.9% of transactions complete processing within $< 1.0$ ms, substantially exceeding Tier-1 banking latency SLAs ($< 35$ ms).

## SI.2 Model Risk Management (MRM) Protocol under Federal Reserve SR 11-7

In compliance with supervisory guidance on Model Risk Management (Federal Reserve SR 11-7 / OCC 2011-12), continuous rolling monitoring operates across three controls: (1) **Quarterly Backtesting:** Non-conformity calibration sets $\mathcal{D}_{\text{cal}}$ are updated on a 90-day rolling window to recalibrate non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$. (2) **Drift Detection & Circuit Breakers:** If the empirical Total Variation drift metric $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{\text{cal}}) > \epsilon_{\text{drift}} = 0.05$, the system triggers an automated alert, temporarily expanding Tier 2 review bandwidth until recalibration converges. (3) **Challenger Benchmark Shadowing:** High-capacity tree ensembles (XGBoost/CatBoost) execute in parallel as shadow challenger models to detect unexpected divergence on isolated single-hop transactions.

# SJ Reproducibility & Open Science Specification

Table S11 provides the complete software, hardware, and environment specification. Table S12 reports the 8-pair zero-shot transfer matrix across frozen backbones, and Table S13 details peak VRAM memory footprint and per-epoch wall-clock timing across benchmark networks. All secondary experimental artifacts—including the 18-parameter hyperparameter configuration grid, the leave-one-out 12-D invariant ablation, and the multi-agent AST grammar schema—are openly available in the project repository: https://anonymous.4open.science/r/Intelligent-AML-9F0A/.

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Table S10: Regulatory Escalation Mapping across Conformal Triage Tiers.

| **Operational Dimension** | **Tier 1 (Escalation Hold)** | **Tier 2 (Boundary Queue)** | **Tier 3 (Automated Clearance)** |
|---|---|---|---|
| Prediction Set $\Gamma(u)$ | $\{1\}$ (Singleton) | $\{0, 1\}$ (Abstain) | $\{0\}$ (Singleton) |
| Risk Category | Confirmed Illicit | Boundary Ambiguity | Confirmed Benign |
| Action Executed | Quarantined Hold | Compliance Queue | Straight-Through Release |
| Statutory Timeframe | Immediate Hold | 72-Hour Review | Real-Time ($< 1$ ms) |
| Regulatory Output | FinCEN Form 111 Dossier | Examiner Audit Log | Cryptographic Clearing Receipt |
| Human Involvement | Mandatory BSA Sign-Off | Full Manual Investigation | Zero (Automated STP) |
| Error Budget | False Discovery $< 1.0\%$ | N/A (Selective Abstention) | False Neg. Leakage $\leq 0.12\%$ |

Table S11: Reproducibility Environment and Software Stack.

| **System Component** | **Specification / Version** |
|---|---|
| Operating System | Ubuntu 22.04.4 LTS (Linux Kernel 6.5) / Windows 11 Pro |
| CPU | AMD Ryzen 9 5900X (12 Cores / 24 Threads @ 3.70 GHz) |
| GPU Accelerator | NVIDIA GeForce RTX 3080 (10 GB GDDR6X) & Tesla T4 (16 GB) |
| System Memory | 64 GB DDR4-3600 ECC / 16 GB Local Edge Profile |
| Python Runtime | Python 3.10.14 (64-bit) |
| Deep Graph Learning | PyTorch 2.2.1+cu121, PyTorch Geometric (PyG) 2.5.2 |
| Tabular Baselines | XGBoost 2.0.3, CatBoost 1.2.3, LightGBM 4.3.0, Scikit-Learn 1.4.2 |
| Streaming Storage | DuckDB 1.0.0, Apache Arrow (PyArrow) 15.0.2, Polars 0.20.15 |
| Hyperparameter Tuning | Optuna 3.6.0 (TPE Sampler), Fixed Seeds $\{42, 101, 2024, 7, 999\}$ |

Table S12: Expanded 8-Pair Zero-Shot Cross-Domain Transfer Matrix under Completely Frozen Neural Backbones ($\eta = 0$). Target-Tuned F1 represents fine-tuning upper bounds where backbones are fully fine-tuned on target training partitions, whereas Zero-Shot F1 evaluates frozen neural backbones ($\eta = 0$) transferring purely canonical 12-D flow invariants.

| Source Dataset | Target Dataset | HGT Baseline F1 | XGBoost Baseline F1 | C-STGB Zero-Shot F1 / PR-AUC | Target-Tuned F1 |
|---|---|---|---|---|---|
| `elliptic_v1` (Bitcoin) | `elliptic_v2` (Multi-Asset) | 21.40% | 64.20% | **82.40%/0.8410** | 99.10% |
| `elliptic_v1` (Bitcoin) | `saml_d` (Multi-Bank) | 14.20% | 58.10% | **76.80%/0.7850** | 93.68% |
| `elliptic_v1` (Bitcoin) | `paysim_extended` (e-Wallet) | 16.80% | 61.40% | **79.20%/0.8120** | 92.40% |
| `paysim_extended` | `ibm_amlsim_hi_med` | 18.50% | 31.40% | **41.80%/0.4320** | 46.85% |
| `paysim_extended` | `saml_d` | 22.10% | 56.80% | **74.50%/0.7620** | 93.68% |
| `dgraphfin` (Dynamic Credit) | `elliptic_v1` | 19.40% | 62.80% | **80.60%/0.8250** | 91.42% |
| `saml_d` | `ibm_amlsim_hi_med` | 15.20% | 28.60% | **39.40%/0.4080** | 46.85% |
| `xblock_eth` | `dgraphfin` | 28.50% | 69.20% | **85.10%/0.8710** | 94.62% |
| **Macro-Average Transfer** | – | **19.51%** | **54.06%** | **69.98%/0.7170** | **82.33%** |

Table S13: Compute Hardware Specifications, Training Wall-Clock Times, and Streaming Latency Metrics across Benchmarks.

| Dataset Identifier | Entities | Peak VRAM | Train s/Epoch | P50 Lat. | P99 Lat. |
|---|---|---|---|---|---|
| `elliptic_v1` | 203,769 | 4.2 GB | 12.4s | 0.38 ms | 1.85 ms |
| `elliptic_v2` | 122,279 | 3.1 GB | 8.6s | 0.35 ms | 1.72 ms |
| `dgraphfin` | 3,064,288 | 15.2 GB | 88.4s | 0.43 ms | 2.14 ms |
| `xblock_eth` | 2,150,000 | 11.2 GB | 62.5s | 0.40 ms | 2.05 ms |
| `saml_d` | 980,000 | 8.4 GB | 38.2s | 0.39 ms | 1.92 ms |
| `paysim_extended` | 1,048,575 | 6.9 GB | 31.0s | 0.36 ms | 1.78 ms |
| `ibm_amlsim_hi_med` | 300,000 | 4.8 GB | 16.8s | 0.37 ms | 1.82 ms |